

Preserved release-gate evidence

Directed modules

controller + sideband
LFSR + error manager
FPGA CSR wrapper

Deterministic UVM

9 named tests
state + transition checks

Production integrated

180 trials; seeds 1701-2105
9 guaranteed scenario classes
398 predictor checks
bypass disabled

Qualified FPGA

wide-top failed fit retained
CSR wrapper fit + internal timing
constraints explicitly bounded

Sideband random

200 trials
seeds 101-505

LFSR random

160 trials
seeds 701-1105

Recovery random

180 trials
seeds 1201-1605

Independent checks

predictor + SVA
explicit sampled counters

All campaigns and implementation reports are retained by one release gate

Accepted release evidence

0 UVM errors | 0 UVM fatals | no reported assertion failures | fixed reproducible seeds